MIL-F-12075D
4 November 1974
SUPERSEDING
MIL-F-12075C
29 July 1966

MILITARY SPECIFICATION

FORGING MACHINE, HAMMER, VERTICAL, ELECTRIC-PNEUMATIC, SELF-CONTAINED AIR SUPPLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers one type of electrically operated, pneumatic forging machine. Machines covered herein have a self-contained air supply system.

* 1.2 Classification. This specification covers forging machines of the the following sizes. The size machine supplied shall be as specified (see 6.2.1).

Size	Ft. Lbs. of Energy Per Stroke (Maximum)
200	1,180
300	1,800
500	3,400
750	5,700
1000	8,800
1500	13,700
2000	18,800
3000	29,500

FSC 3446

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2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments	Occupational Safety and Health Administration Standards

NATIONAL BUREAU OF STANDARDS (NBS) HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

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(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Ave., NW, Washington, DC 20005.)

3. REQUIREMENTS

* 3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract. The approved sample shall be used as the standard for all production items manufactured to fulfill the terms of the contract.

* 3.2 Design. The machine shall be new and one of the manufacturer's current models. All parts subject to wear shall be easily accessible for adjustment, replacement, and repair.

* 3.2.1 Controls. All operating controls shall be located convenient to the operator at his normal work station. Controls shall include means for starting and stopping the machine and other necessary controlling devices selectable by the operator in controlling the machine's operating functions.

* 3.2.2 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards which can be guarded without substantial interference to operation of the machine. All guards shall be removable to provide access to guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.4).

* 3.2.3 Lubrication. The machine shall be equipped with a means of lubrication for all bearings (except sealed, permanently lubricated type), all mating gears, ways and sliding parts. An integral forced-feed system shall lubricate the ram and compressor cylinders. All oil reservoirs shall have filler strainers and a means for determining oil level.

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Manually operated lubrication systems for ways and remote areas shall have control handles mounted in an accessible location. All oil holes, grease fittings and filler caps shall be easily accessible.

* 3.2.4 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensional and tolerance standards, which will permit installation of replacement parts without modification to the part or equipment.

* 3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

* 3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging or filling with solder or paste shall be used for reclaiming any defective part.

* 3.3.2 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

* 3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

* 3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

* 3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and in accordance with NBS H28.

* 3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

* 3.3.8 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

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* 3.4 Components. All machines covered herein shall consist essentially of a base and frame, drive system, ram and cylinder, compressor cylinder, anvil, dies, their related electrical, control and lubricating systems, and such other components required to meet the requirements specified herein.

* 3.4.1 Frame. The machine shall have a "C" type frame consisting essentially of a base, column, and ram cylinder. The ram cylinder shall be integral with the column, and the column and base may be either a single integral structure or two separate structures rigidly and accurately bolted together. The base shall incorporate, support, or encircle an anvil conforming to 3.4.2. The frame shall support or enclose the drive system, the compressor, and all other components specified herein. The frame shall have adequate strength and rigidity to withstand all stress produced when operating the machine at its maximum capacity without permanent deformation or premature failure. The throat depth of the frame shall conform to the requirements of Table I. The frame shall have an adequate number of covered openings of ample size to facilitate servicing, maintenance, and adjustment of all components contained therein. The base shall have means for securely fastening the machine to the floor or a foundation.

* 3.4.2 Anvil. The anvil shall be a casting of iron or steel with an anvil to ram weight ratio of not less than 15:1. On size 200 machines, the anvil shall be cast integrally with the base and frame of the machine. On all other size machines, the anvil may be either integrally cast with the base and frame, supported by the base, or free standing, passing through the base.

* 3.4.2.1 Anvil cap. The anvil cap shall be securely attached to the anvil in such a manner to permit its removal from the anvil block without disturbing the hammer.

* 3.4.3 Drive system. The machine shall have a mechanical type drive system conforming to 3.4.3.1 thru 3.4.3.1.4.

* 3.4.3.1 Mechanical type drive. The mechanical type drive shall include a drive motor, transmission, flywheel, drive shaft, compressor piston and connecting rod, and ram and cylinder unit.

* 3.4.3.1.1 Drive motor. The drive motor shall conform to the requirements of 3.4.8.1 and Table I.

* 3.4.3.1.2 Transmission. The transmission shall be of the gear or V-belt type. V-belt systems shall be capable of transmitting the required power without belt slippage, and shall have provisions for adjusting belt tension.

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* 3.4.3.1.3 Flywheel. The flywheel shall be a balanced assembly having adequate mass and rotational speed and the capability of supplying supplementary power to the motor drive in maintaining a constant speed, within 10 percent of no load speed, while providing the energy specified in Table I.

* 3.4.3.1.4 Drive shaft. The drive shaft shall be capable of transmitting power developed by the drive train and shall be of adequate size and strength to prevent deflection when subjected to maximum torque. The drive shaft shall be mounted on replaceable bronze bushings or anti-friction bearings of a size and type suitable for the application.

* 3.4.4 Compressor piston and connecting rod. The compressor piston and cylinder shall be sized to provide the number of blows per minute specified in Table I. The piston shall be fitted with replaceable piston rings. The connecting rod shall be a steel forging and shall be mounted on replaceable bronze bushings or anti-friction bearings of a size and type suitable for the application. The connecting rod and piston shall be driven by crank or eccentric mounted on the drive shaft.

* 3.4.5 Slide and ram cylinder unit. The slide and ram cylinder unit shall provide reciprocating motion for the forging operation. The ram and cylinder unit shall consist of a double acting piston with the piston connected to the ram. The cylinder shall be sealed against leakage by replaceable sealing devices. The ram shall be of solid or box-type construction, with reciprocating action guided by a slide piston-cylinder or gib-way arrangement. The guide or gib shall be adjustable and replaceable. Provisions shall be made for cushioning the ram at the upper end of the stroke. The bottom face of the ram shall be dovetailed or grooved to hold the ram die.

* 3.4.6 Dies. The forging machine shall be furnished with forging dies for the ram and anvil cap. Dies shall have straight flat faces for plain forgings and shall be made of forged steel of die block quality. Dies shall be securely attached to the anvil cap and ram by steel taper keys.

* 3.4.7 Operating controls. The forging machine shall be equipped for hand lever and foot treadle operation for controlling blows of the ram in accordance with 3.6. Control valves shall provide constant number of blows, variable blow intensity, and squeeze or hold down operations. An unloading or start valve shall be provided on machines larger than Size 500 to allow motor starting without compression buildup. Electric motor controls shall conform to the requirements of 3.4.8.

* 3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall be in accordance with JIC Electrical Standards for General Purpose Machine Tools. An identified terminal shall be pro-

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vided for grounding the machine when installed. The terminal shall be suitable for connecting the size grounding conductor specified in Table 15-1 of JIC Standards. Each machine shall accept all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit fed through a fused safety disconnect switch. Unless otherwise specified (see 6.2.1), the machine shall be initially wired for operation on 230 volts. The control conductors between components which are separable for shipment of the machine shall be connected through one or more separable plug connectors.

* 3.4.8.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings of the main drive motor shall conform to the requirements of Table I. Unless otherwise specified (see 6.2.1), each motor shall meet the requirements of a dripproof enclosure.

* 3.5 Sizes and characteristics. Unless otherwise specified (see 6.2.1), the sizes and characteristics of the machine shall be not less than those specified in Table I.

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TABLE I

Sizes and Characteristics

Characteristics	SIZE							
	200	300	500	750	1000	1500	2000	3000
Ft. Lbs of Energy	1,180	1,800	3,400	5,700	8,800	13,700	18,800	29,500
Blows Per Minute of Ram	195	180	120	125	115	100	90	85
Stroke Length of Ram, Inches	12	14	15	22	27	32	32	38
Die Surface Size, Inches	2-3/4x6	3-1/2x7-7/8	3-1/2x9	5x10	5x11	6x13-1/2	7x14	8x16
Motor HP	7-1/2	10	15	25	40	60	75	125
Throat Depth, Inches	10	11	14	16-3/4	18-1/2	28	28	32

NOTE: Size designation is the nominal weight, in pounds, of the ram and all attached parts that fall with it. Requirements are not less than those shown.

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3.6 Performance. The forging machine shall conform to the performance characteristics, capacities, and general dimensions of Table I. The hammer shall be capable of performing general types of smith forging operations. Ram control shall be by hand lever and foot treadle linked together to allow operation by either control on sizes 200 through 500. Foot treadle is not mandatory on sizes greater than 500. The machine shall have provisions for clamping work between the dies by a squeezing or holding action.

* 3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

One set of hand tools normally furnished by the manufacturer

* 3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

* 3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

* 3.10 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall be in accordance with Table II of MIL-STD-461 for Class IIB equipment.

* 3.11 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

* 3.12 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricants

Viscosity

Military or Federal Specification covering required lubricant

* 3.13 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate

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shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Capacity, ram, foot pounds of energy

Power input characteristics and rating

Date of manufacture

Contract Number or Order Number

Federal Stock Number, National Stock Number or Plant Equipment Code

U. S.

* 3.14 Technical data. Technical data shall be furnished as specified (see 6.2.2).

* 3.15 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1 and 6.6), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the First Article.

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* 4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, test in 4.5.1, and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

* 4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.7 through 3.15.

4.5 Tests.

* 4.5.1 Operational test. The machine shall be operated at no load for a minimum of 30 minutes, during which time the machine drive and operating controls shall be actuated and observed to determine compliance with the requirements specified herein.

* 4.5.2 Performance test. The intensity of the hammer blow (ft. lbs of energy per stroke) shall be calculated as follows:

$$E = \frac{WV^2}{2g}$$

Where: E = Forging energy per blow (ft.-lb.)

V = Velocity (fps) of impacting weight (speed of falling ram from the topmost to the bottom positions with no workpiece between dies).

W = Impacting weight (lb.) plus force (lb.) applied to impacting weight (weight of falling ram plus force pounds of pneumatic pressure applied to top of falling ram).

g = Acceleration of gravity = 32.2 ft. per second^2.

Failure to conform to energy requirements of Table I shall be cause for rejection. (Prospective purchasers are referred to 6.6 for additional information concerning this performance test.)

* 4.5.3 Electromagnetic interference tests. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.10 in accordance with Table II of MIL-STD-461.

* 4.6 Inspection of preparation for delivery. Preservation-packaging, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

* 5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2.1).

* 5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B, or C as specified (see 6.2.1).

* 5.3 Marking. Marking shall be in accordance with MIL-STD-129. Additional marking shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for use in general types of forging operations. The machines are intended for prototype, short run or production operation.

* 6.2 Ordering data.

* 6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Size of machine required (see 1.2 and Table I).

c. First Article approval, if required (see 3.1).

d. Specify exceptions and additional safety and health requirements (see 3.2.2 and 6.4).

e. Painting, if different (see 3.3.5).

f. Electrical equipment, if different (see 3.4.8).

g. Voltage requirements, if different (see 3.4.8).

h. Type of enclosure for motors, if different (see 3.4.8.1).

i. Sizes and characteristics, if different (see 3.5).

j. Standard equipment, if different (see 3.7).

k. Optional equipment, as specified and fully described (see 3.8).

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l. List of repair parts required (see 3.9).

m. Electromagnetic interference control, if required (see 3.10).

n. Type fungus proofing, if required (see 3.11).

o. Lubrication chart, if different (see 3.12).

p. Nameplate, if different (see 3.13).

q. First Article inspection, if different; If manufacturers certification is acceptable in lieu of performing the test in 4.5.2, so state (see 4.2 and 6.5).

r. Quality conformance inspection, if different (see 4.3).

s. Electromagnetic interference control test requirements, if different (see 4.5.3).

t. Required level of preservation-packaging (see 5.1).

u. Required level of packing (see 5.2).

v. Additonal marking, if required (see 5.3).

* 6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

* 6.4 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

* 6.5 Mercury prohibited. When the use of mercury or its compounds in the machine and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

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* 6.6 Manufacturer's certification in lieu of performance testing. Paragraph 4.2 requires that the First Article inspection, when required, include the performance test in 4.5.2 to prove that the forging energy of the machine meets the requirements of Table I. Although the energy may be readily calculated from the simple equation shown in 4.5.2, determination of the "velocity of impacting weight" requires special laboratory type equipment not normally owned by manufacturers of forging machines. The cost of determining this velocity could considerably increase the cost of a contract for purchasing these machines. To minimize contract cost, consideration should be given to accepting the machine manufacturer's certification that the energy expended by the hammer conforms to the requirements of Table I, in lieu of performing the test in 4.5.2. Such a certification can be based on the original machine design calculations, industry accepted empirical equations, or similar energy tests performed in the past on a similar machine. If such a certification is acceptable, the procuring agency should so state under the option in 6.2.1, q.

* 6.7 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes and additions to the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - WC
Navy - SH, YD, AS
Air Force - 84
DSA - GS

Preparing Activity:
DSA - IP

Project Number:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   [ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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